

Calculus Practice — Gradient Descent

7 questions, mixing hand-computation, closed-form analysis, and conceptual reasoning. Try all of them before checking the answers below.

Practice Questions

Q1 Let $f(x) = x^2 - 6x + 5$. Starting at $x_0 = 0$ with learning rate $\alpha = 0.1$, compute x_1 and x_2 by hand using the gradient descent update rule $x_{n+1} = x_n - \alpha f'(x_n)$.

Q2 Let $f(x, y) = 2x^2 + y^2$. Starting at the point $(1, -2)$ with learning rate $\alpha = 0.2$, compute the point after **one** gradient descent step: $(x, y) \leftarrow (x, y) - \alpha \nabla f(x, y)$.

Q3 Consider the family of functions $f(x) = ax^2$ for a constant $a > 0$, and gradient descent with a constant learning rate α , starting at x_0 . Show that after n steps:

$$x_n = x_0(1 - 2a\alpha)^n$$

(Hint: find the relationship between x_{n+1} and x_n first, then recognize the pattern.)

Q4 Using the result from Q3, describe the long-run behavior of x_n in each of these three cases: (a) $0 < \alpha < \frac{1}{a}$, (b) $\alpha = \frac{1}{a}$, (c) $\alpha > \frac{1}{a}$.

Q5 Still for $f(x) = ax^2$: there is one specific learning rate α^* for which gradient descent lands **exactly** on the minimum after just a single step, regardless of the starting point x_0 . Find α^* , and explain what makes it special compared to the range of learning rates you found in Q4.

Q6 The gradient descent update moves in the direction of the *negative* gradient, $-\nabla f$, rather than the gradient itself. Using the fact that ∇f points in the direction of steepest increase of f , explain why this is the right choice for minimizing f .

Q7 Recall the function $E(x) = x^4 - 4x^3 - 2x^2 + 12x + 3$, which has two global minima (at $x = -1$ and $x = 3$) separated by a local maximum at $x = 1$. Explain why gradient descent starting at $x_0 = 0.5$ and gradient descent starting at $x_0 = 1.5$ can converge to two completely different minima, even though the two starting points are close together.

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Answers & Solutions

A1. $f'(x) = 2x - 6$.

$$x_1 = x_0 - \alpha f'(x_0) = 0 - 0.1(2(0) - 6) = 0 - 0.1(-6) = 0.6$$

$$x_2 = x_1 - \alpha f'(x_1) = 0.6 - 0.1(2(0.6) - 6) = 0.6 - 0.1(-4.8) = 0.6 + 0.48 = 1.08$$

(The true minimum is at $x = 3$, so both steps are moving steadily in the right direction.)

A2. $\nabla f(x, y) = (4x, 2y)$. At $(1, -2)$: $\nabla f(1, -2) = (4, -4)$.

$$(x, y) \leftarrow (1, -2) - 0.2(4, -4) = (1 - 0.8, -2 + 0.8) = (0.2, -1.2)$$

A3. $f'(x) = 2ax$, so the update rule is:

$$x_{n+1} = x_n - \alpha(2ax_n) = x_n(1 - 2a\alpha)$$

This means each step just multiplies the previous value by the constant factor $(1 - 2a\alpha)$. Starting from x_0 and applying this n times gives:

$$x_n = x_0(1 - 2a\alpha)^n$$

A4. Let $r = 1 - 2a\alpha$, so $x_n = x_0 r^n$.

- (a) For $0 < \alpha < \frac{1}{a}$, we get $-1 < r < 1$, so $r^n \rightarrow 0$ and $x_n \rightarrow 0$ — gradient descent **converges** to the minimum.
- (b) For $\alpha = \frac{1}{a}$, we get $r = -1$, so $x_n = x_0(-1)^n$ — the iterate **oscillates forever** between x_0 and $-x_0$, never converging and never diverging.
- (c) For $\alpha > \frac{1}{a}$, we get $|r| > 1$, so $|x_n| \rightarrow \infty$ — gradient descent **diverges** (oscillating with growing magnitude, since r is negative once $\alpha > \frac{1}{2a}$).

A5. Setting $r = 1 - 2a\alpha = 0$ gives:

$$\alpha^* = \frac{1}{2a}$$

What makes this special: it's exactly the **midpoint** of the convergence range $(0, \frac{1}{a})$ from Q4, and it's the unique learning rate that makes the multiplying factor r equal to zero — so the very first step lands exactly on $x = 0$, the minimum, no matter how far away x_0 started. Every other learning rate in the convergence range approaches the minimum only gradually.

A6. ∇f points toward the direction in which f increases the fastest. Since the goal of gradient descent is to *decrease* f , moving a small step in the opposite direction, $-\nabla f$, is — at least locally — the direction in which f decreases the fastest. Repeating this many times steadily walks f downhill toward a minimum.

A7. $x = 1$ is a local maximum, so it acts as a “ridge” separating the function into two basins: everything to its left flows toward the minimum at $x = -1$, and everything to its right flows toward the minimum at $x = 3$. Since $0.5 < 1$ and $1.5 > 1$, the two starting points fall on opposite sides of that ridge — even though they're close together, gradient descent always moves *away* from the local maximum (since the gradient points uphill toward it), so each point slides down into the basin it happens to start in.